Android 3.0 offers a new hardware-accelerated OpenGL renderer that gives a performance boost to many common graphics operations for applications running in the Android framework. When the renderer is enabled, most operations in Canvas, Paint, Xfermode, ColorFilter, Shader, and Camera are accelerated. Developers can control how hardware-acceleration is applied at every level, from enabling it globally in an application to enabling it in specific Activities and Views inside the application.

12.10.3 Renderscript 3D graphics engine

Renderscript is a runtime 3D framework that provides both an API for building 3D scenes as well as a special, platform-independent shader language for maximum performance. Using Renderscript, you can accelerate graphics operations and data processing. Renderscript is an ideal way to create high-performance 3D effects for applications, wallpapers, carousels, and more.

12.11 Support for multicore processor architectures

Android 3.0 is the first version of the platform designed to run on either single or multicore processor architectures. A variety of changes in the Dalvik VM, Bionic library, and elsewhere add support for symmetric multiprocessing in multicore environments. These optimizations can benefit all applications, even those that are single-threaded. For example, with two active cores, a single-threaded application might still see a performance boost if the Dalvik garbage collector runs on the second core. The system will arrange for this automatically.

12.14 Rich multimedia and connectivity

HTTP Live streaming

Applications can now pass an M3U playlist URL to the media framework to begin an HTTP Live streaming session. The media framework supports most of the HTTP Live streaming specification, including adaptive bit rate.

12.14.1 Pluggable DRM framework

Android 3.0 includes an extensible DRM framework that lets applications manage protected content according to a variety of DRM mechanisms that may be available on the device. For application developers, the framework API offers an consistent, unified API that simplifies the management of protected content, regardless of the underlying DRM engines.

12.14.2 Digital media file transfer